

Generation Interconnection Request System Impact Study Report

For Queue Project Q14-0417

Walnut 345 kV
July 20, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0417. The Interconnection Customer proposes a 210.6 MW wind facility interconnecting at the Walnut 345 kV substation in the Pennsylvania area. The request was received on December 27, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 214 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 15 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$79,897,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Cortland / Berry Rd 138.0 kV Line: 165.5% loading (189 MVA) following the outage of the Walnut / Clear Springs 345.0 kV Line; DFAX = 12.5%.

For the loss of Clear Springs Substation / Walnut 345.0 kV, the Berry Rd Substation / Florence 138.0 kV loads to 162.7% of its 211 MVA rating. Project DFAX: 12.5%.

Fiberboard / Warrick 138.0 kV L: 158.5% loading (209 MVA) following the outage of the Clear Springs 500-345 kV Transformer Bank; DFAX = 5.5%.

For the loss of Walnut / Clear Springs 345.0 kV, the Florence / W Hamlin 138.0 kV loads to 158.4% of its 222 MVA rating. Project DFAX: 12.5%.

W Hamlin Substation / Boyds Mill Hydro 138.0 kV: 156.3% loading (232 MVA) following the outage of the Clear Sprngs / Walnut Substation 345.0 kV Line; DFAX = 12.5%.

The Mears Corners / Boyds Mill Hydro 138.0 kV is overloaded to 153.5% of its 246 MVA emergency rating for the loss of the Walnut / Clear Springs 345.0 kV. The project contributes 12.5% DFAX (26.2 MW) to this overload.

Shawnee / Latigo 138.0 kV: 151.7% loading (207 MVA) following the outage of the Clear Springs 500/345kV Autotransformer; DFAX = 5.5%.

For the loss of Walnut / Clear Springs 345.0 kV, the Walnut / Mears Corners 138.0 kV loads to 151.4% of its 262 MVA rating. Project DFAX: 12.5%.

The Yards Crk Substation / Latigo 138.0 kV is overloaded to 138.3% of its 220 MVA emergency rating for the loss of the Clear Springs 500 kV/345 kV XFMR. The project contributes 5.5% DFAX (11.5 MW) to this overload.

The Warrick / Shawnee 138.0 kV is overloaded to 136.3% of its 231 MVA emergency rating for the loss of the Clear Springs 500/345kV Transformer. The project contributes 5.5% DFAX (11.5 MW) to this overload.

For the loss of Clear Springs / Walnut 345.0 kV, the Cortland / Lesbia 138.0 kV L loads to 132.4% of its 220 MVA rating. Project DFAX: 12.5%.

Walut / Clear Springs 345.0 kV: 120.2% loading (1,101 MVA) following the outage of the Tombstone / Bixby Hill 500.0 kV; DFAX = 45.5%.

The Bar Beach 345-138 kV Transformer Bank is overloaded to 110.6% of its 365 MVA emergency rating for the loss of the Clear Springs Substation 500 kV/345 kV Autotransformer. The project contributes 10.0% DFAX (21.1 MW) to this overload.

Belding / Lesbia 138.0 kV L: 109.1% loading (243 MVA) following the outage of the Clear Springs / Walnut 345.0 kV; DFAX = 12.5%.

Clear Springs 500/345 kV Transformer Bank: 108.2% loading (1,228 MVA) following the outage of the Alderton / Clear Springs 345.0 kV; DFAX = 30.2%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Cortland / Berry Rd Substation 138.0 kV Line	Rebuild line with higher-capacity conductor	\$1,061,000	100% (\$1,061,000)
Berry Rd Substation / Florence 138.0 kV	Rebuild line with higher-capacity conductor	\$5,561,000	100% (\$5,561,000)
Fiberboard / Warrick 138.0 kV L	Rebuild line with higher-capacity conductor	\$6,595,000	100% (\$6,595,000)
Florence / W Hamlin 138.0 kV	Rebuild line with higher-capacity conductor	\$12,865,000	100% (\$12,865,000)
W Hamlin Substation / Boyds Mill Hydro 138.0 kV	Rebuild line with higher-capacity conductor	\$1,528,000	100% (\$1,528,000)
Mears Corners / Boyds Mill Hydro 138.0 kV	Rebuild line with higher-capacity conductor	\$3,662,000	100% (\$3,662,000)
Shawnee / Latigo 138.0 kV	Rebuild line with higher-capacity conductor	\$5,109,000	100% (\$5,109,000)
Mears Corners / Walnut 138.0 kV	Rebuild line with higher-capacity conductor	\$4,012,000	100% (\$4,012,000)
Yards Crk Substation / Latigo 138.0 kV	Rebuild line with higher-capacity conductor	\$3,510,000	100% (\$3,510,000)
Shawnee / Warrick 138.0 kV	Rebuild line with higher-capacity conductor	\$3,722,000	100% (\$3,722,000)
Cortland / Lesbia 138.0 kV L	Rebuild line with higher-capacity conductor	\$2,605,000	100% (\$2,605,000)
Clear Springs / Walnut 345.0 kV	Rebuild line with higher-capacity conductor	\$36,474,000	34% (\$12,481,000)
Bar Beach 345-138 kV Transformer Bank	Replace transformer with higher-rated unit	\$12,339,000	100% (\$12,339,000)
Lesbia / Belding 138.0 kV	Reconductor line	\$318,000	100% (\$318,000)
Clear Springs 500 kV/345 kV Transformer	Replace transformer with higher-rated unit	\$25,226,000	18% (\$4,529,000)
	Total allocated to project		\$79,897,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.